



STUDYBUDDY AI

MINI STARTUP PLAN

"Scan Your Textbook. Master Your Exam."

An Artificial Intelligence-driven study platform enabling high-retention self-learning through dynamic syllabus extraction and targeted mock evaluations.

Subject: Artificial Intelligence (Subject Code 417)

Level: Class 10 Internal Assessment (Entrepreneurial Skills)

Academic Session: 2026-2027

Board: CBSE Curriculum Standard (NCERT Syllabus)

TABLE OF CONTENTS

1. Executive Summary	Page 3
2. Problem Scoping & 4Ws Canvas	Page 3
3. The Proposed Solution (Business Idea)	Page 4
4. Market Analysis & Competitive Edge	Page 5
5. Target Audience & User Personas	Page 5
6. Financial & Operational Budget (INR ₹)	Page 6
7. Technical Framework & AI pipeline	Page 7
8. Ethical Safety & Privacy Policies	Page 7
9. Strategic Roadmap & Conclusion	Page 8



Figure 1.1: Official StudyBuddy AI Launch Advertisement & Campaign Creative



1. EXECUTIVE SUMMARY

StudyBuddy AI is a mobile-first, AI-powered learning application designed to improve how school students in India study independently. The product leverages Optical Character Recognition (OCR) and high-speed Large Language Models (LLMs) to make learning from dense textbooks fast and engaging.

By taking a photo of any CBSE NCERT textbook page, students instantly receive:

- **Structured Bullet summaries** highlighting primary keywords and definitions.
- **A custom-generated 10-question quiz** matching the exact style (MCQs, assertion-reasoning) of their CBSE board exams.

Priced at an affordable **₹99/month**, StudyBuddy AI targets a large gap between complex web-based chatbots and high-cost private coaching, providing middle-class Indian parents with an effective and affordable homework support tool.



2. PROBLEM SCOPING (THE 4WS CANVAS)

In accordance with the standard **AI Project Cycle**, the problem is scoped across four specific domains:

Who (Who is affected by the problem?)

School students in Classes 9 to 12 who face high academic stress and dense workloads. Secondly, parents who want to support their child's education but cannot afford expensive private tutors.

What (What is the problem?)

School textbooks are extremely dense. Students struggle to extract key points quickly during revision. Existing revision tools (like search engines or generic chatbots) are either slow, distracting, or do not align with the specific CBSE board syllabus.

Where (Context/Setting?)

This problem is most common during self-study hours at home, late-night exam preparations, or quick review sessions in school buses immediately before a test.

Why (Why solve it?)

Providing students with structured revision aids and active practice quizzes reduces exam anxiety, improves retention of information, and saves parents thousands of rupees on private tutoring.

[4ws Problem Scoping Canvas]

— WHO: Classes 9-12 Students & worried parents

— WHAT: Overwhelming textbook text, generic search results, expensive tutoring

- |— WHERE: Self-study hours, exam-eve preparation, home learning environments
- |— WHY: Standardizes exam practice, reduces test anxiety, provides affordable aid

3. THE PROPOSED SOLUTION (BUSINESS IDEA)

StudyBuddy AI provides a seamless study flow that translates physical text into structured study notes in less than 5 seconds.



Core Application Features:

- **Intelligent OCR Engine:** High-precision text capture that extracts readable content even from low-light photos or curved book pages.
- **Curriculum-Aware Summaries:** The system automatically aligns the explanation layout strictly with CBSE NCERT learning outcomes.
- **10-Question Syllabus Quizzes:** Instant mock quizzes featuring Multiple Choice Questions (MCQs), fill-in-the-blanks, and assertion-reasoning questions.
- **Progress Metrics Dashboard:** Interactive streaks, XP tracking, and difficulty metrics to keep students motivated.



4. MARKET ANALYSIS & COMPETITIVE EDGE

While standard chatbots like ChatGPT and Google Gemini are highly advanced, they are not optimized for K-12 school students.

Why General Chatbots Fail Younger Students:

- 1. **Complex Prompting Required:** Students must write detailed prompts to receive age-appropriate responses, which is slow and frustrating.
- 2. **Syllabus Mismatch:** General LLMs draw from global databases and often include deleted topics or materials outside the specific national boards.
- 3. **Distraction Risks:** Open chat systems do not contain children-safe educational filters, raising parental concerns.

Competitive Comparison Matrix

Feature	General AI Chatbots	Traditional EdTech Packs	StudyBuddy AI
Usage Speed	Slow (Requires text prompts)	Slow (Hours of video lectures)	Instant (1-click photo scan)
Syllabus Focus	Generic / General Knowledge	Static/Pre-recorded courses	Dynamic & CBSE-Tailored
Safety Focus	None (Unrestricted web access)	High (Closed app)	High (Educational filters only)
Affordability	Free or ₹1,999/month	₹2,000 - ₹5,000/month	₹99/month (Premium)

5. TARGET AUDIENCE & USER PERSONAS

Our audience profile consists of two distinct segments:

User Profile A: Priya Sharma (Class 10 Student)

Priya is studying for her CBSE science boards. She is overwhelmed by the length of the chemistry chapters and wants to quickly check her knowledge before a school test without scrolling through hours of YouTube videos. StudyBuddy AI provides her with instant summaries and quick practice questions.

User Profile B: Rajesh Patel (Parent)

Rajesh wants his son to perform well in his studies, but local private coaching institutes charge over ₹3,000 per month. He purchases the ₹99/month StudyBuddy Pro plan, which provides an affordable way for his son to practice high-quality educational assessments at home.

6. FINANCIAL & OPERATIONAL BUDGET (INR ₹)

StudyBuddy AI uses a **Freemium Business Model** to balance wide user reach with healthy startup profitability.

- Free Plan:** Includes 3 book scans per day, standard summaries, and basic text outputs.
- Premium Plan (Pro):** Priced at **₹99 per month**, offering unlimited scans, CBSE-specific styling, PDF exports, custom quizzes, and text-to-speech audio.

Estimated Monthly Operational Expenses (Year 1 Budget)

Expense Category	Operational Description	Cost in INR (₹)
AI API Tokens	Inference fees for LLM queries (Gemini 1.5 Flash API)	₹25,000
Cloud Servers	AWS/Google Cloud database hosting & static website files	₹12,000
Marketing Campaign	Targeted social media promotions to student communities	₹15,000
Maintenance & Support	Customer service, standard app maintenance, and bug fixes	₹20,000
UPI Gateway Fees	Payment transaction charges (2%) & buffer costs	₹8,000
TOTAL MONTHLY EXPENSES		₹80,000

Profit Projection & Break-Even Analysis

Assuming a target of **10,000 active free users** and a conservative **1,000 paying premium subscribers** (10% conversion) in the first year:

- **Gross Monthly Revenue:** $1,000 \text{ subscribers} \times ₹99 = ₹99,000$
- **Total Monthly Cost:** ₹80,000
- **Net Monthly Profit:** $₹99,000 - ₹80,000 = ₹19,000$

The break-even point represents the number of premium subscribers required to cover all operating costs:

$$\text{Break-Even Point} = ₹80,000 / ₹99 \approx 809 \text{ Premium Subscribers}$$

7. TECHNICAL FRAMEWORK & AI PIPELINE

StudyBuddy AI uses a modern, lightweight tech stack to ensure fast responses and low costs:

- 1. Frontend Interface:** Built with **Flutter** (for cross-platform iOS & Android mobile performance) and **HTML5/CSS3** for web views.
- 2. Text Extraction (Data Acquisition):** **Google Cloud Vision OCR API** for converting textbook images into readable text.
- 3. AI Logic & Summaries (Modelling):** **Google Gemini 1.5 Flash API**. This model is selected for its high speed (responses under 2 seconds) and low token costs, making our ₹99/month price point highly sustainable.
- 4. Formatting Layer:** Predefined prompt templates format the output into structured JSON, ensuring consistent layouts for study materials.

8. ETHICAL SAFETY & PRIVACY POLICIES

An outstanding school project report must address safety and ethical considerations:

- **Data Privacy (DPDP Act & COPPA compliance):** StudyBuddy AI does not collect or sell personal student data. Uploaded textbook photos are processed in memory and deleted from servers immediately after text extraction.
- **Homework Integrity:** To prevent cheating on assignments, the app implements a "Homework Mode" policy. During standard school hours (8:00 AM to 2:00 PM), the app provides conceptual hints rather than direct answers for complex math and science homework.
- **Accuracy & Mitigating Hallucinations:** The generative LLMs are restricted strictly to the context of the scanned textbook page, preventing the system from generating incorrect facts or external material.

9. STRATEGIC ROADMAP & CONCLUSION

StudyBuddy AI demonstrates how artificial intelligence can be used to make quality revision tools affordable and accessible to school students. It transforms self-study from passive reading into active, personalized learning.

Future Development Phases:

- **Multilingual Support:** Translating scanned English textbook pages into Hindi, Tamil, Bengali, or Marathi to support CBSE students who prefer learning key terminology in their native mother tongue.
- **Speech-to-Study Voice Assistant:** Allowing visually impaired students to listen to interactive audio summaries and answer study quizzes vocally.
- **Collaborative Class Quizzes:** A safe lobby where friends can challenge each other on quizzes generated directly from their shared school syllabus.